

TECUCI, Romania

WMO: 152650

Lat: 45.8417N

Lon: 27.4089E

Elev: 61

StdP: 100.59

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-10.4	-15.6	1.0	-11.6	-13.5	1.2	-9.5	12.9	-4.9	11.3	-5.4	2.4	340	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.8	33.1	21.1	31.7	20.7	30.3	20.4	22.9	30.0	21.9	29.1	21.2	28.2	2.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	15.3	26.2	19.6	14.5	24.6	18.9	13.8	23.8	67.7	30.2	64.3	29.4	61.7	28.6	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.3	7.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-2.8	1.8	5.7	11.0	17.3	21.4	22.3	22.7	17.9	11.0	6.0	0.2		
	DBStd	9.56	4.40	4.84	4.60	4.36	3.21	2.86	2.50	3.01	4.10	4.06	4.26	4.45		
	HDD10.0	1289	396	231	144	42	2	0	0	0	1	37	132	304		
	HDD18.3	3019	654	463	393	222	57	7	2	4	58	228	369	563		
	CDD10.0	1749	0	1	10	72	228	343	383	394	237	68	13	0		
	CDD18.3	437	0	0	0	1	24	100	127	139	45	1	0	0		
	CDH23.3	4304	0	0	3	29	297	912	1179	1421	451	12	0	0		
	CDH26.7	1573	0	0	0	2	62	311	438	622	139	0	0	0		
(14) Wind	WSAvg	3.0	3.1	3.1	3.6	3.5	3.0	3.0	2.7	2.9	2.6	2.5	2.7	3.0		
(15) Precipitation	PrecAvg	540	25	24	29	45	61	73	60	50	48	52	34	36		
	PrecMax	787	54	80	69	91	138	176	171	157	141	207	109	118		
	PrecMin	313	2	3	3	16	8	11	9	12	5	2	1	1		
	PrecStd	130	17	18	17	23	36	43	38	35	34	43	26	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	16.7	22.3	26.2	31.1	33.7	34.2	34.8	32.0	25.3	20.0	12.8		
		MCWB	5.3	10.4	12.5	16.6	19.3	22.5	21.9	21.7	19.2	17.5	14.4	9.4		
	2%	DB	6.1	13.1	18.1	23.9	28.5	31.7	32.4	33.5	30.0	22.6	17.1	10.2		
		MCWB	4.4	8.7	10.6	15.1	17.7	21.5	21.4	20.8	19.1	15.8	12.7	7.7		
	5%	DB	4.5	10.8	15.6	21.4	26.6	30.1	30.9	32.0	28.2	20.2	14.1	7.8		
		MCWB	3.1	7.1	9.4	13.8	17.4	21.0	21.1	20.5	18.0	14.5	10.9	6.0		
	10%	DB	2.7	8.4	13.1	19.3	24.9	28.4	29.2	30.4	25.9	17.9	12.0	5.5		
		MCWB	1.6	5.8	7.9	12.6	16.5	20.1	20.5	20.1	17.5	13.2	10.0	3.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	12.0	13.4	17.2	20.3	24.3	23.7	24.5	21.3	18.4	14.7	10.5		
		MCDB	7.9	15.7	21.0	25.0	28.8	31.0	31.6	33.0	28.5	23.8	19.0	11.6		
	2%	WB	4.7	8.9	11.2	15.7	19.2	23.0	22.5	22.5	19.9	16.5	13.1	8.2		
		MCDB	5.9	11.9	17.4	22.6	26.1	29.4	29.9	30.3	26.7	21.0	16.5	10.1		
	5%	WB	3.2	7.3	9.7	14.5	18.1	21.8	21.6	21.7	19.1	15.2	11.7	6.2		
		MCDB	4.3	9.9	14.4	20.3	24.5	28.1	28.8	29.3	25.7	19.0	13.8	7.9		
	10%	WB	1.8	6.1	8.6	13.2	17.0	20.8	20.9	20.9	18.1	14.0	10.1	4.1		
		MCDB	2.7	8.2	12.3	18.3	22.9	27.0	28.1	27.9	24.7	17.0	11.8	5.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	7.6	9.9	11.0	12.0	11.4	11.7	12.8	11.9	10.0	7.1	5.7		
		MCDBR	7.7	12.3	15.5	15.5	15.3	13.8	14.7	15.4	15.7	14.1	11.4	8.8		
		MCWBR	5.8	8.1	8.6	8.2	6.3	5.5	5.3	5.3	6.3	7.6	6.9	6.4		
	5% WB	MCDBR	6.9	10.6	13.7	14.7	13.2	12.6	13.2	13.8	14.0	12.7	10.2	8.6		
		MCWBR	5.4	7.5	8.2	7.8	6.0	5.4	5.2	5.1	6.1	7.2	6.5	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.362	0.397	0.457	0.429	0.428	0.447	0.460	0.418	0.383	0.357	0.331		
	taud		2.312	2.245	2.198	2.055	2.217	2.256	2.211	2.162	2.243	2.321	2.344	2.326		
	Ebn at Noon		753	797	821	801	841	842	819	790	791	765	716	708		
	Edh at Noon		79	103	124	156	138	133	138	139	118	95	77	71		
(44) All-Sky Solar Radiation	RadAvg		1.29	2.09	3.25	4.43	5.61	5.93	6.03	5.46	3.94	2.56	1.39	1.07		
	RadStd		0.16	0.25	0.33	0.44	0.47	0.48	0.47	0.39	0.37	0.29	0.13	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	+0.57	N/A	N/A	N/A	N/A	N/A	-115	-155	+73	+29

Nomenclature: See separate page